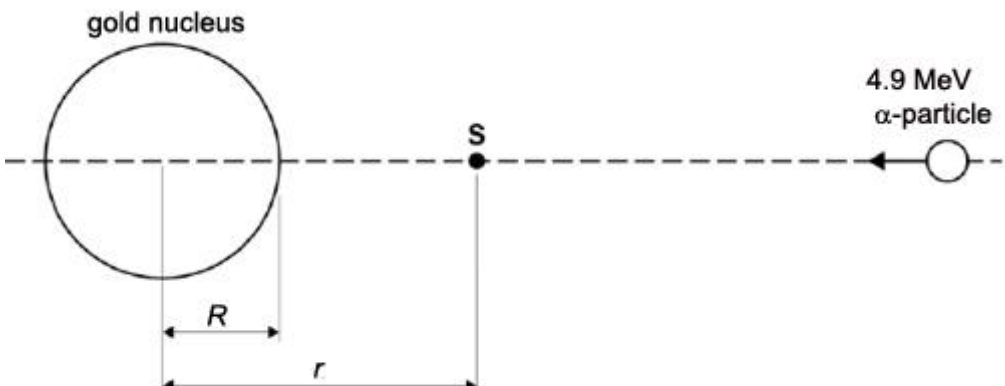
17 An α-particle with an initial kinetic energy of 4.9 MeV is directed towards the centre of a gold nucleus of radius *R* which contains 79 protons. The radius *R* of the gold nucleus is found to be 1.4×10^{-14} m.

The α-particle is brought to rest at point S, a distance *r* from the centre of the nucleus as shown in the figure.



What is the distance *r* of the α-particle to the nucleus?

A 1.4×10^{-14} m **B** 4.6×10^{-14} m **C** 2.2×10^{-7} m **D** 4.4×10^{-7}m